Final Presentation Design-for-Test

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Spring 2026

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Timeline

1. Introduction to DFT concepts and familiarization with Tessent shell
2. Gain knowledge about Tessent shell commands
3. Tessent scan and ATPG modules and exam
4. Tessent ATPG with TestKompress module and exam
5. Implementation in RV321 5-stage RISC-V processor
6. Poster and final presentation

Introduction

What is DFT?

- DFT is the specific hardware design toward faults identifications
- DFT enables and promotes diagnose of manufacturing defects

Why is DFT important?

- DFT allows early fault detection, boosted faults coverage, simple debugging, improvement in product reliability, and decrease in cost

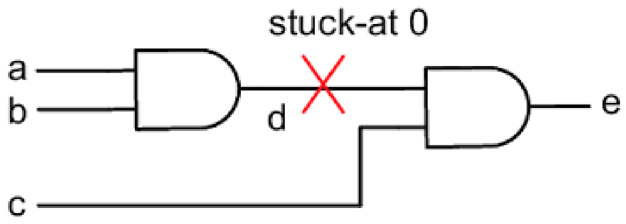
Tessent

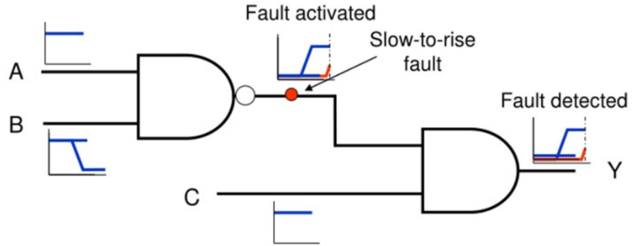
- Tessent is the Siemens EDA test suite, utilized to insert and manage test patterns in integrated circuits
- Used in the design phase, Tessent follows a specific workflow: EDT/OCC insertion, synthesis, scan insertion, ATPG, and simulation

Manufacturing Defects

- Design defects can be detected during the design process and if unnoticed can be present in all fabricated devices

- Manufacturing defects happen on a device/set basis due to fabrication issues. We model them with fault models:





Stuck-At Faults: A node is modeled as permanently tied to logic 0 or logic 1. Detected by applying input vectors that sensitize the fault and propagate its effect to an observable output.

Transition Faults: A node is modeled as switching too slowly to meet timing. Detected using at-speed test patterns where the OCC issues a fast capture pulse after a slow shift sequence.

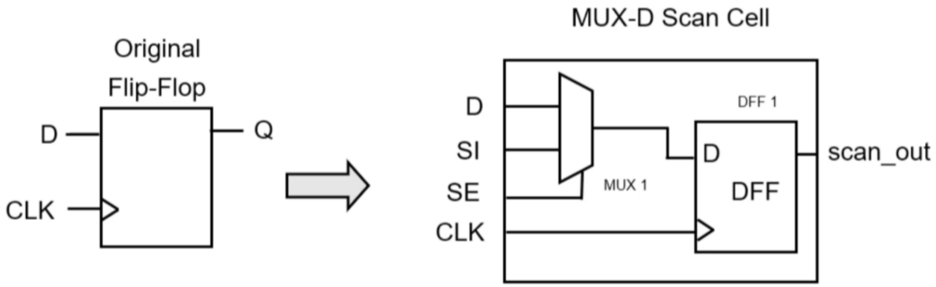
The ATE (Automatic Testing Equipment)

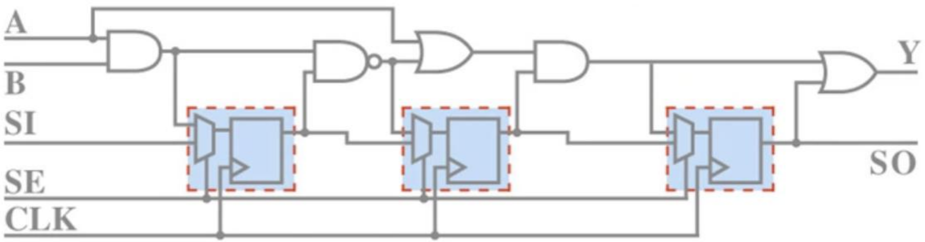
- It is the machine responsible for setting up and colleting the test data.
- Wafers go into a dedicated space in the "arm" of the ATE, a probe card acts as the "adapter" for the pins.
- ATE machines are extremely expensive, so reducing test time is an important consideration in reducing chip cost

The Advantest V93000

Scan Cells & The Capture Cycle

- Scan cells are Flip-Flops with a MUX added to its input
- SE (Scan Enable) changes FF input to SI (Scan Input) instead of D.
  - This allows data to be almost directly inserted into each FF for testing, improving controllability
- Scan Cells can be serialized, creating Scan Chains
- The capture cycle:
  - Load/Unload: SE ON, Load Testing values into Scan Cell through SI
  - Capture: SE OFF, move data through circuit in functional mode to perform normal operations
  - Load/Unload: SE ON, Unload values by shifting them through scan chain, Load new values





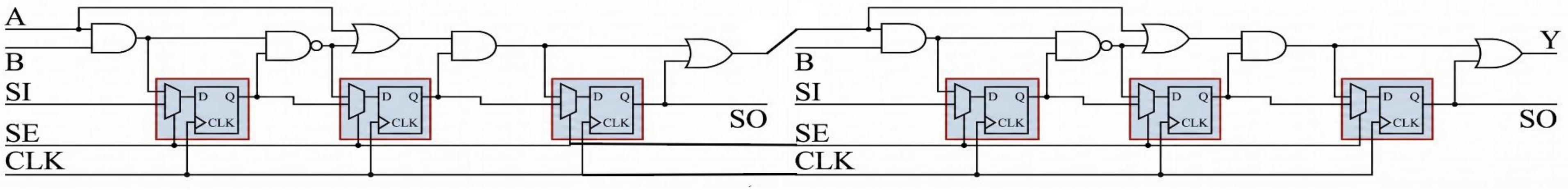
Serial and Parallel Scan Chains

- A series of Scan Cells is a Serial Scan Chain; you can break a Serial Chain into Parallel Chains

- Each Parallel Chain needs an independent SI and SO, reducing test time (less shifts due to less Scan Cells in series) but increasing the number of pins and the power consumptions

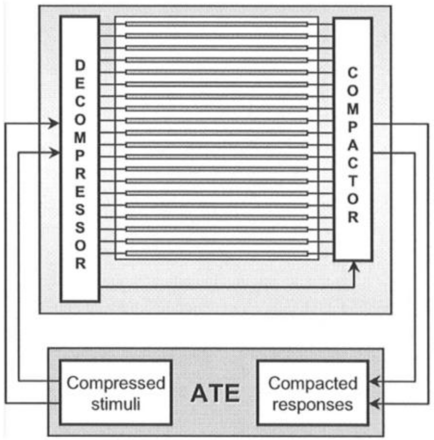
- In Serial Chains, you need to shift inputs into the Scall Cells, and to shift the outputs out of the cells, increasing the test time, but decreasing the number of pins.

- Ideally you want to find the equilibrium of not having too long chains nor too many testing pins



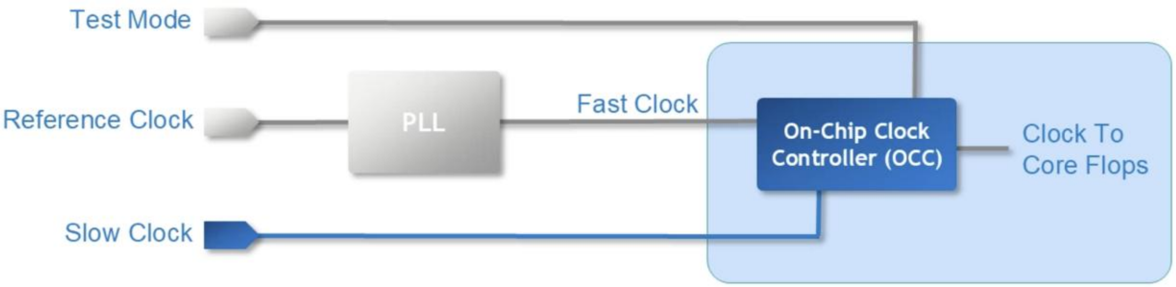
The EDT (Embedded Deterministic Test)

- More advanced designs require more higher volumes of input data, but usually only 1% to 5% of bits are really needed
- ATE sends compressed testcases that are decompressed in an on-chip decompressor with a ring generator and phase shifter
- EDT supports more parallel scan chains while having few inputs from the ATE
- Compactor masks SCs to remove X blocking
- Achieves high compression with low area addition and without affecting the design performance



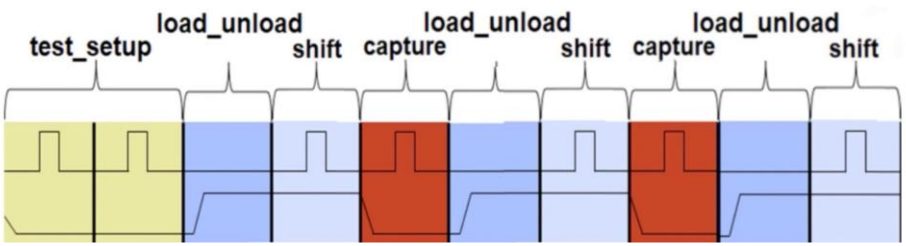
The OCC (On-Chip Clock Controller)

- ATE is usually much slower than chip clock (MHz vs GHz), which could lead to tests missing timing faults
- Slow Clock (the ATE clock) is used for load/unload of test patterns & shift cycles
- Fast Clock (the chip/PLL clock) is used for the capture cycle
- OCC is needed to be able to switch between clock modes, while also avoiding considerable delays to fast clock

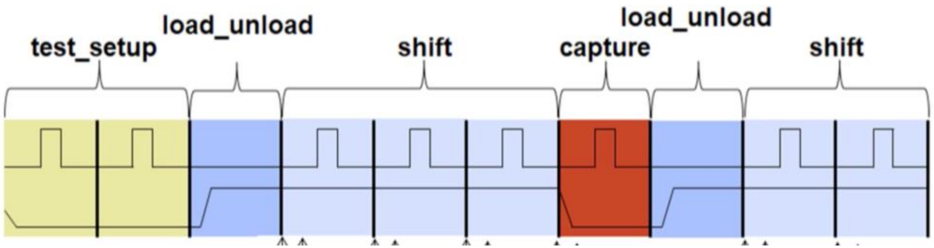


Parallel vs Serial Testbenches

- Parallel testbenches excite multiple individual scan cells at the same time, while serial requires the propagation of values through the scan chain
- Parallel is faster, but it can only detect problems during capture
- Serial excites EDT and the full scan path, and detects issues within capture and shift



Parallel Testbench



Serial Testbench

Implementation - Flow

EDT/OCC Insertion
Insert EDT controller and OCC into RTL.

Tessent Synthesis
Synthesize RTL with DFT logic.

Scan Insertion
Replace FFs with scan FFs, stitch scan chains.

Stuck-At ATPG
Generate stuck-at patterns.

Transition ATPG
Generate transition patterns with OCC fast clock.

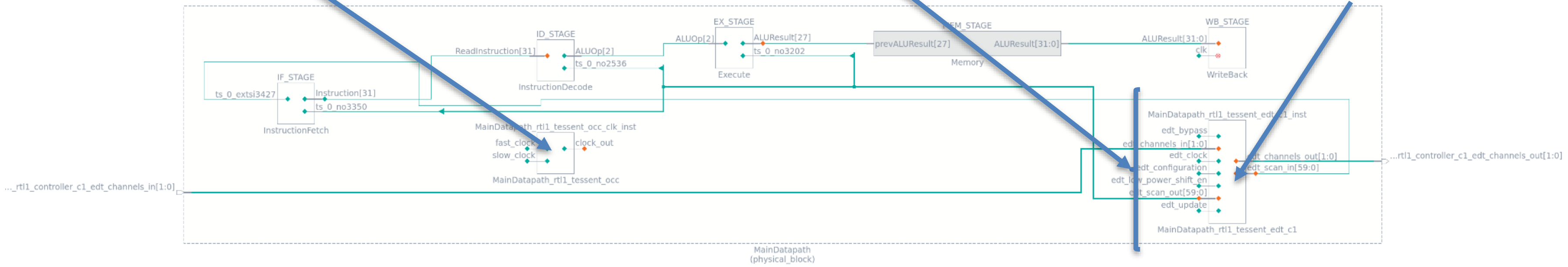
Model Sim
Waveform view of simulated generated patterns.

Overview Hardware Diagram

On-Chip Clock Controller: Multiplexes between slow shift clock and fast clock. Enables at-speed capture for transition fault testing.

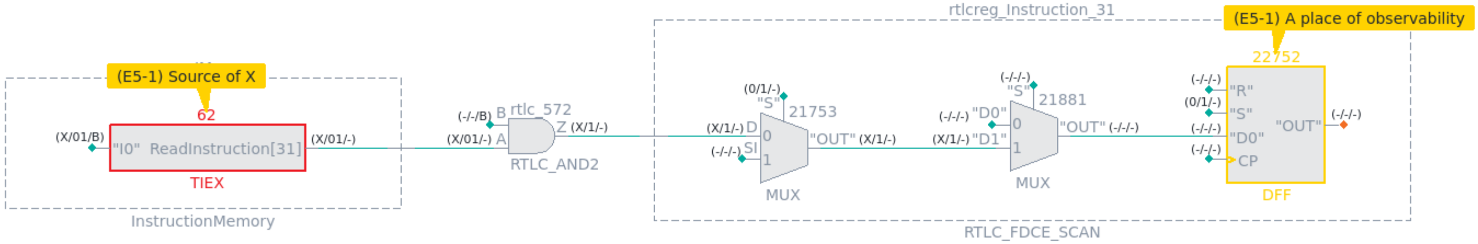
EDT Control signals: scan_en gates shift vs. capture; edt_clock/edt_update; edt_bypass routes channels directly to chains for uncompressed testing; edt_low_power_shift_en gates chain segments during shift; edt_configuration selects the operating mode.

EDT Controller: The EDT controller sits between the ATE and 60 internal scan chains. A decompressor expands 2 input channels (edt_channels_in[1:0]) into (edt_scan_in[59:0]), and a compactor compresses 60 chain responses back into 2 output channels (edt_channels_out[1:0]).



Violations

E5 Violation: When the application places constrained states on constrained pins and binary states on PIs scan cells, X states must not propagate to an observable point. Failure to satisfy this rule will result in the risk of X states propagating to an observable point.

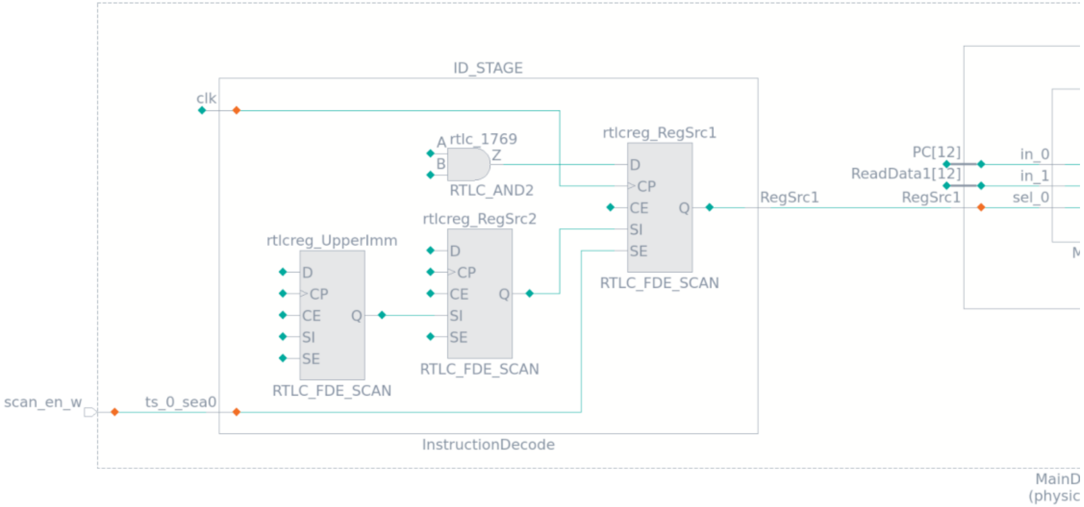



```
add_nofaults -module { \
    InstructionMemory \
    DataMemory \
}
```

DFT Implementation Summary

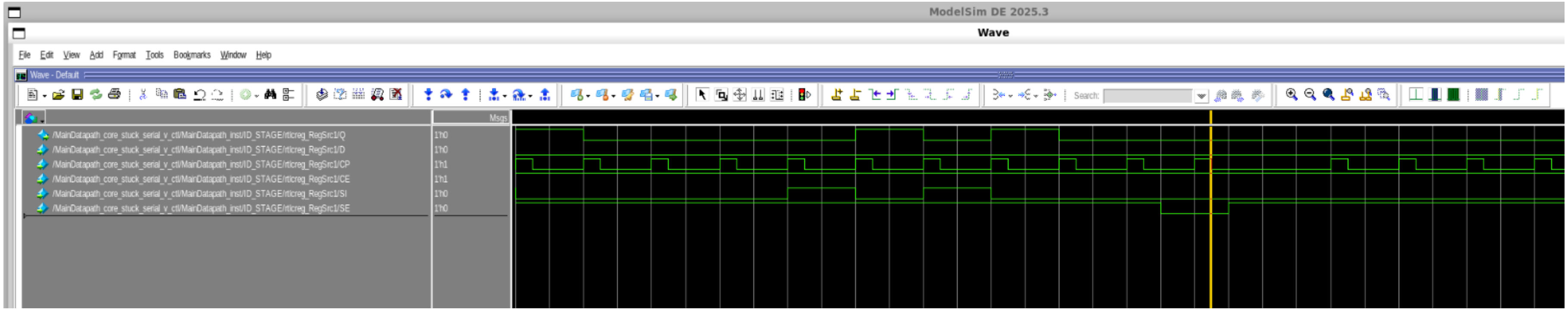
Parameter	Value
Total memory elements	2,178
Scan cells	2,008
Non-scan	170
Scan chains	60
Chain length	33-34
Lockup latches	60
EDT channels	2 in/2 out
Total gates	24,656
Primary inputs	18
Primary outputs	310

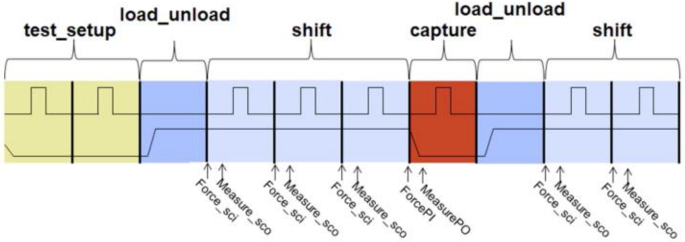
Example: Scan Cell



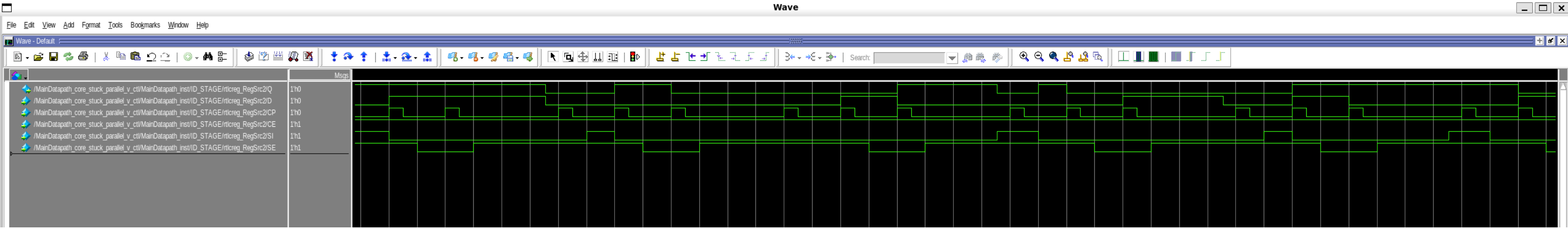

```
model RTLC_FDE_SCAN
  (Q, D, CP, CE,
   SI, SE)
(
  model_source = verilog_module;
  rtlc_type = flop;
  nonscan_model = RTLC_FDE;

  input (D) ( )
  input (CP) ( posedge_clock; )
  input (CE) ( )
  input (SI) ( scan_in; )
  input (SE) ( scan_enable; )
  output (Q) ( scan_out; )
  (
    instance = RTLC_MUX2 mux1 (D_in, D, SI, SE);
    instance = RTLC_OR2 or1 (CE_in, CE, SE);
    instance = RTLC_PRIM_DFFE g1 (Q, CE_in, D_in, CP, mlc_n0, mlc_n1);
    primitive = _tie0 mlc_tie0_1 (mlc_n0);
    primitive = _tie0 mlc_tie0_2 (mlc_n1);
  )
) // end model RTLC_FDE_SCAN
```

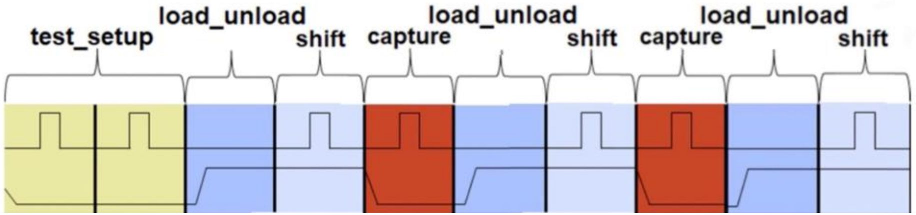
Shift then Capture Sequence – Serial Pattern



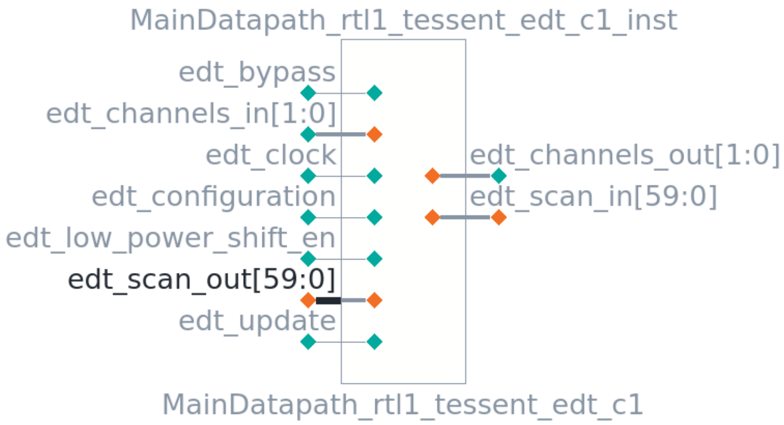


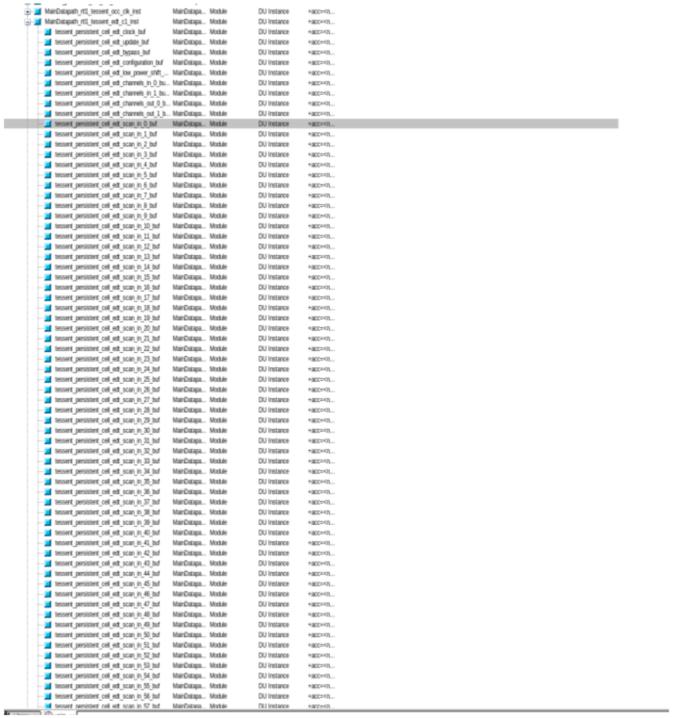
Shift then Capture Sequence – Parallel Pattern



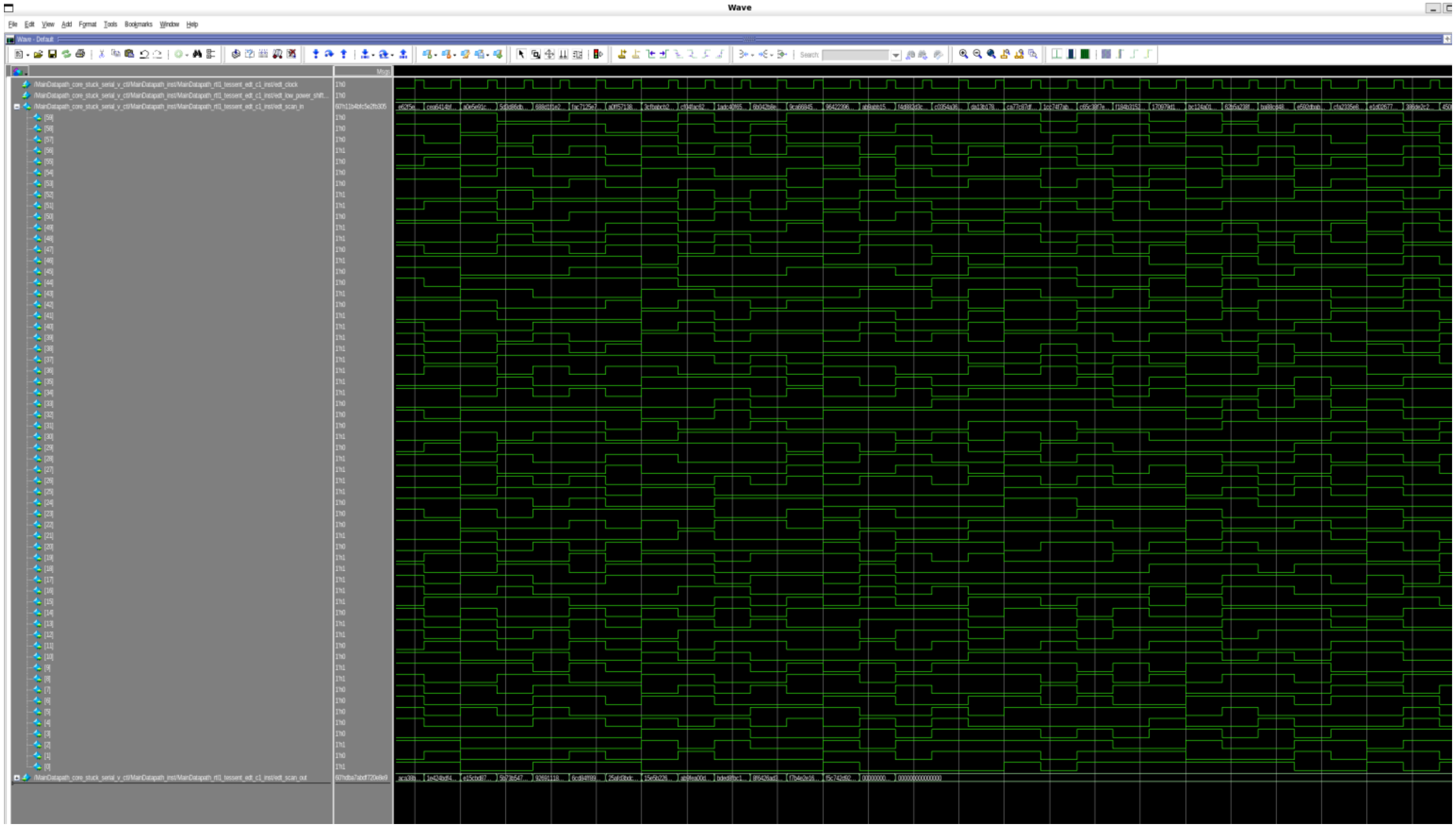


Scan Chains

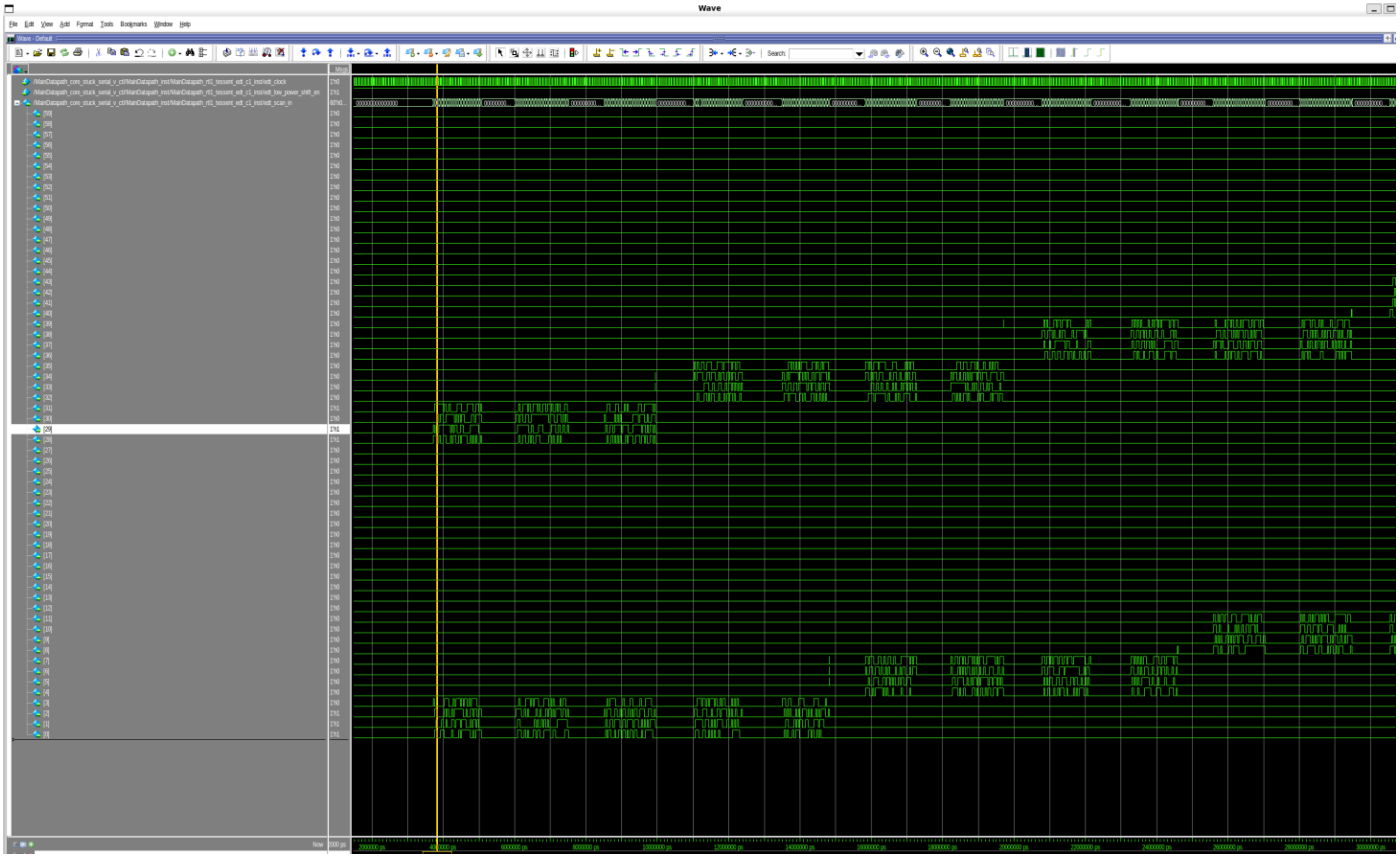

MainDatapath_rtl1_tessent_edt_c1_inst
edt_bypass
edt_channels_in[1:0]
edt_clock
edt_configuration
edt_low_power_shift_en
edt_scan_out[59:0]
edt_update
edt_channels_out[1:0]
edt_scan_in[59:0]
MainDatapath_rtl1_tessent_edt_c1



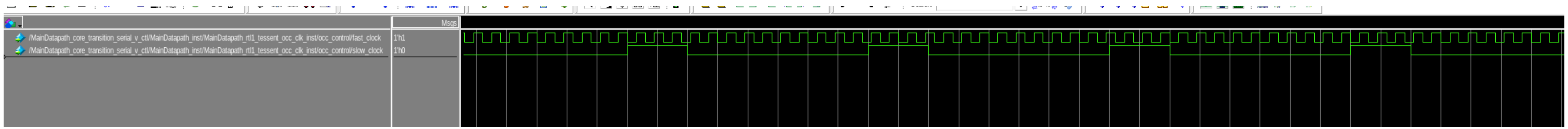
Low Power Disabled

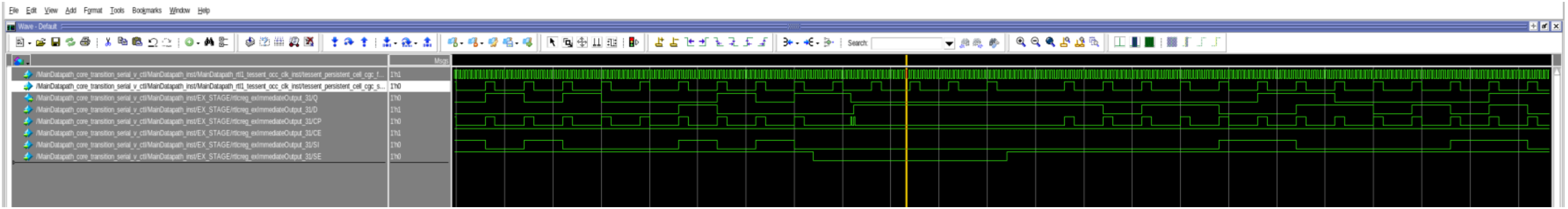


Low Power Enabled – 15%



Transition Faults and OCC





```
set_fault_type TRansition
set_external_capture_options -pll_cycles 5 [lindex [get_timeplate_list] 0]
```

ATPG results across configurations

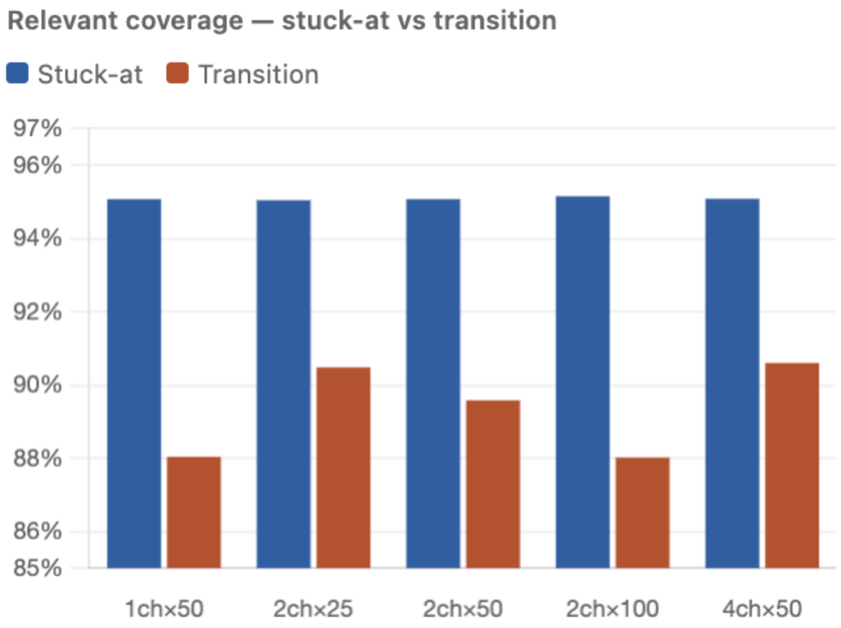
Configuration	Chain Length	Total Coverage	Relevant Coverage	Patterns	Scan Volume
1ch x 50 chains	40-41	89.49%	95.07%	329	662.6K
2ch x 25 chains	80-81	90.26%	95.04%	317	666.5K
2ch x 50 chains	40-41	87.15%	95.07%	327	661.2K
2ch x 100 chains	20-21	81.97%	95.15%	329	665.3K
4ch x 50 chains	40-41	86.75%	95.08%	339	718.1K

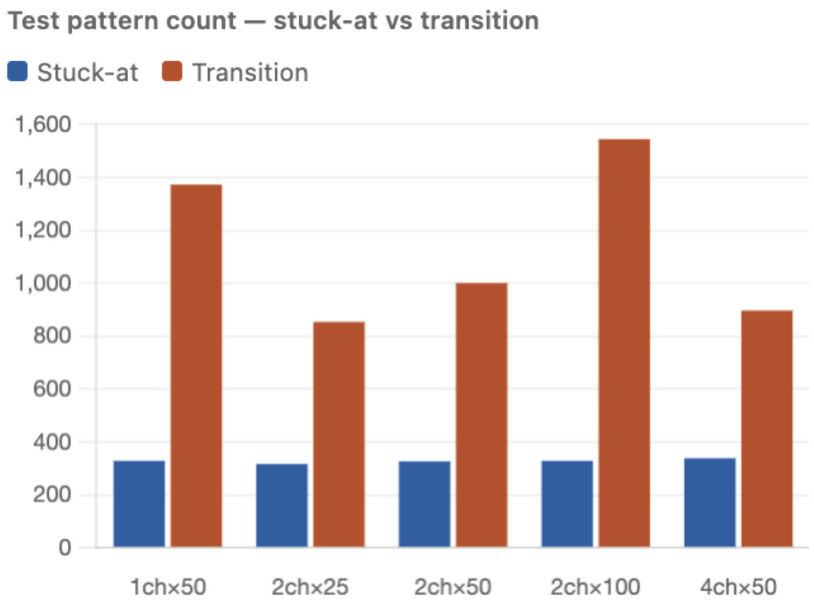
Stuck at fault result summary

Configuration	Total Coverage	Relevant Coverage	Patterns	EAB aborts	ATPG effectiveness
1ch x 50 chains	87.90%	88.04%	1,373	662.6K	662.6K
2ch x 25 chains	90.40%	90.48%	854	666.5K	666.5K
2ch x 50 chains	89.44%	89.58%	1,001	661.2K	661.2K
2ch x 100 chains	87.74%	88.02%	1,545	665.3K	665.3K
4ch x 50 chains	90.46%	90.60%	897	718.1K	718.1K

Transition fault result summary

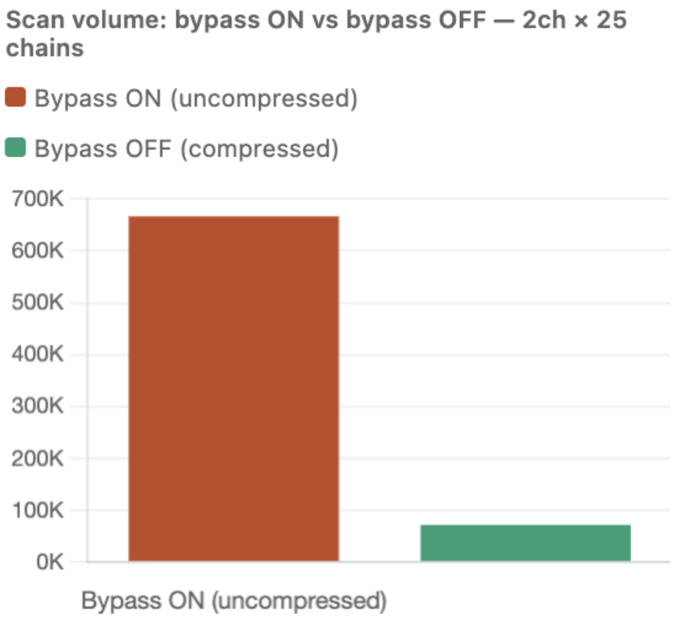
Stuck-at vs Transition



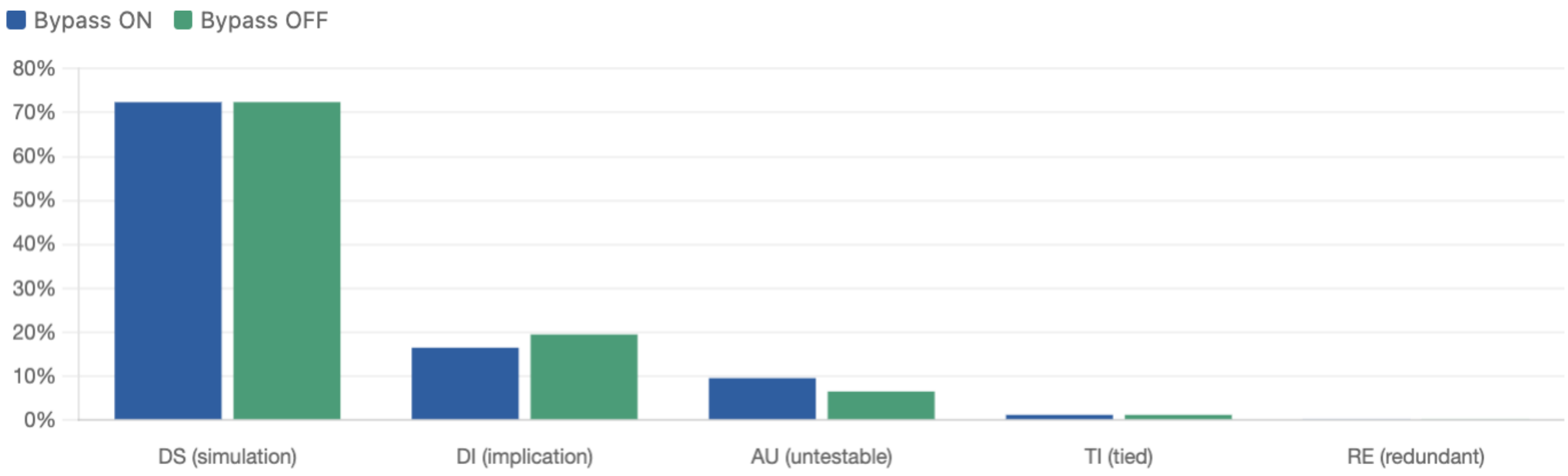


Bypass Vs No-Bypass

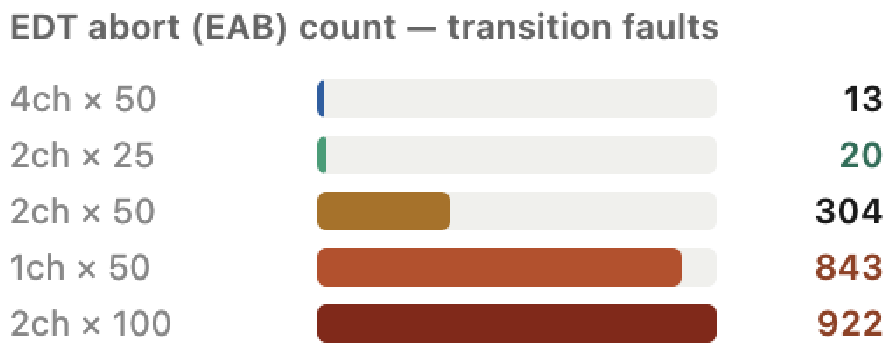
Metric	Bypass ON (uncompressed)	Bypass OFF (EDT compressed)
Scan volume	666.5K cell loads	71.6K cell loads
Shift cycles	1,048 cycles	81 + 15 cycles
Test patterns	317	357
Total coverage	90.26%	93.35%
Relevant coverage	95.04%	95.20%
DI (implication)%	16.49%	19.51%



Fault Classification



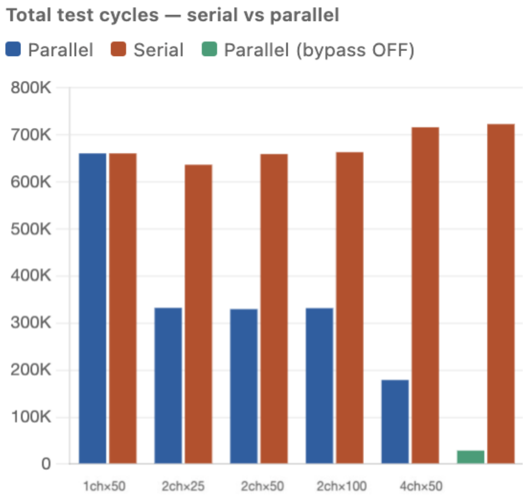
EDT abort (EAB) count



- EAB count shows how often Tessent fails to compress a transition-fault pattern.
- 4ch × 50 and 2ch × 25 configurations have the lowest abort counts
- 2ch × 100 has the worst result.
- This suggests that very short chains hurt compression efficiency

Serial vs Parallel (Stuck-at)

Configuration	Chain	Longest chain	Serial shift cycles	Total Serial Cycles	Parallel shift cycles	Total Parallel Cycles	Parallel speedup
1ch x 50 chains	1	2,008	2,008	660,632	2,008	660,632	1.0x
2ch x 25 chains	2	1,048	2,008	6636,536	1,048	332,216	1.9x
2ch x 50 chains	2	1,008	2,016	659,232	1,008	329,616	2.0x
2ch x 100 chains	2	1,008	2,016	663,264	1,008	331,632	2.0x
4ch x 50 chains	4	528	2,112	716,208	528	178,992	4.0x
2ch x 25 chains (Bypass off)	25	81	2,025	722,925	81	28,917	25.0x



Low Power Disabled vs Enabled

Switching Threshold Percentage	Avg WSA (%)	Discarded Patterns	Test Coverage (%)
25	20.9	10	95.52
20	10.09	69	95.47
15	12.9	3,015	82.43

Challenges & Solutions

Familiarize with content

Siemens EDA educational platform modules and clarifying weekly meetings

Install Tessent Shell

Guided setup with user license key provided by Pratt OIT

Familiarize with Tessent Shell commands

Execute the labs, gaining proficiency with Tessent commands and workflow

Find Open source – RISC-V CPU

Found a simple 5-stage pipelined processor on GitHub – RV321 processor

Deal with memory when generating test patterns

To avoid a lack of specific complex memory tasks, we added black boxes to the memory components

Conclusion & Future Work

- There is a severe need for hardware testing optimization
- Tessent both automates and optimizes this process
- We implemented DFT infrastructure, generating ATPG patterns, black-boxing memory, and analyzing the results
- Achieved ~95% fault coverage for stuck-at faults and ~90% for transition faults

- Test memory
- Implementing Memory BIST (MBIST)
- Generate memory test algorithms
- Integrating MBIST into a processor
- Validate memory test operation

Acknowledgements:

We thank Siemens EDA for providing access to the Tessent toolchain and training, and Linus Tauro, Haitham Nofal, and Jonathan Fan for their technical guidance throughout the project. We also thank Dr. Daniel Sorin for inviting us to take the class.